



**WEST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED**

(A GOVT. OF WEST BENGAL ENTERPRISE)

**TECHNICAL SPECIFICATION FOR 13M/11M HOT ROLLED PARALLEL FLANGE STEEL SECTIONS**

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## 1. SCOPE:

This specification covers design, manufacture, galvanizing, testing and supply of 13m/11M long WPB 150X150X36.98 RS JOIST.

## 2. STANDARDS:

The RS JOISTS shall comply with the requirements of IS 12778:2004 and IS 2062:2011 with Grade Designation: E250 (Fe 410 W) and Quality: "A" in respect of "dimensions and section properties" and steel specifications respectively.

### 3. DIMENSIONS AND SECTIONAL PROPERTIES:

| Designation  | Mass  | Area            | Depth | Width | Thickness |      | Root radius | Moment of Inertia |                 | Radius of Gyration |                | Sectional modulus |                 |
|--------------|-------|-----------------|-------|-------|-----------|------|-------------|-------------------|-----------------|--------------------|----------------|-------------------|-----------------|
|              |       |                 |       |       | Web       |      |             | I <sub>x</sub>    | I <sub>y</sub>  | r <sub>x</sub>     | r <sub>y</sub> | Z <sub>x</sub>    | Z <sub>y</sub>  |
|              |       |                 |       |       | M         | a    |             |                   |                 |                    |                |                   |                 |
|              | kg/m  | Cm <sup>2</sup> | mm    | mm    | mm        | mm   | mm          | Cm <sup>4</sup>   | Cm <sup>4</sup> | Cm                 | Cm             | Cm <sup>3</sup>   | Cm <sup>3</sup> |
| WPB 150X150X | 36.98 | 47.1            | 162   | 154   | 8         | 11.5 | 8           | 2210              | 706.2           | 6.85               | 3.87           | 273.2             | 91.5            |

#### 4. ROLLING AND CUTTING TOLERANCE:

#### 4.1 Dimensional Tolerance

| Designation          | Depth | Width | Thickness |        |
|----------------------|-------|-------|-----------|--------|
|                      |       |       | Web       | Flange |
| WPB<br>150X150X36.98 | D     | B     | t         | T      |
|                      | mm    | mm    | mm        | mm     |
|                      | 7/3   | 7/3   | 7/1       | 7/1.5  |

Other dimensional tolerance will be in conformity to IS 12779:1989.

## 4.2 Mass Tolerance

The tolerance on mass per metre of individual section shall be  $\pm 4$  percent of the specified mass i.e. 36.98Kg/m. Mass tolerances for lot however shall be within  $\pm 3$  percent of the specified values.

### 4.3 Cutting Tolerance

Cutting tolerance shall be within +100, -0 mm from the specified values.

5. EMBOSSING ON EACH JOIST:

Following Distinct non-erasable embossing to be made on each RS JOIST:

- Name & Logo of the manufacturer.
- ISI mark.
- Designation of steel and Size of the JOIST

# GENERAL TECHNICAL PARTICULARS

FOR

WPB 150X150X36.98

## 1. PARALLEL FLANGE STEEL SECTIONS:

### 1.1 Dimensions and Properties:

| Sl.no. | Particulars                                                                                 | WPB<br>150X150X36.98                          | Manufacturer's Data |
|--------|---------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------|
| 1      | Length of Joist in Mtr.                                                                     | 13Mtr/11Mtr                                   |                     |
| 2      | Weight Kg/m                                                                                 | 36.98kg/m                                     |                     |
| 3      | Depth (D) of section in mm                                                                  | 162mm                                         |                     |
| 4      | Width (W) of section in mm                                                                  | 154mm                                         |                     |
| 5      | Thickness of flange (T) in mm                                                               | 11.5mm                                        |                     |
| 6      | Thickness of web (t) in mm                                                                  | 8mm                                           |                     |
| 7      | Root radius (R) in mm                                                                       | 8mm                                           |                     |
| 8      | Moment of Inertia<br>I <sub>x</sub> (cm <sup>4</sup> )<br>I <sub>y</sub> (cm <sup>4</sup> ) | 2210 cm <sup>4</sup><br>706.2 cm <sup>4</sup> |                     |
| 9      | Radius of Gyration (cm)<br>r <sub>x</sub><br>r <sub>y</sub>                                 | 6.85cm<br>3.87cm                              |                     |
| 10     | Sectional modulus<br>Z <sub>x</sub> (cm <sup>3</sup> )<br>Z <sub>y</sub> (cm <sup>3</sup> ) | 273.2 cm <sup>3</sup><br>91.5 cm <sup>3</sup> |                     |

### 1.2 Mechanical properties:

| Sl.no. | Test                                                                                                                                                      | Requirement as per IS<br>2062:2011 with Grade<br>Designation: E250 (Fe 410<br>W) and Quality: "A" | Manufacturer's<br>Data |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------|
| A      | Tensile Test:<br>i. Yield Stress (MPa)<br>ii. Tensile Strength (MPa)<br>iii. Percentage Elongation A, Min<br>at Gauge Length, L <sub>o</sub> =5.65        | Min250<br><br>Min 410<br>23                                                                       |                        |
| B      | Bend test:<br>The test piece at room temperature<br>shall withstand bending without<br>cracking through 180° to an internal<br>diameter not greater than: | Internal Bend Diameter:<br>2X Thickness of test piece.                                            |                        |

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1.3 Chemical properties:

| Sl.no. | Test                       | Requirement as per IS 2062:2011 with Grade Designation: E250 (Fe 410 W) and Quality: "A" | Manufacturer's Data |
|--------|----------------------------|------------------------------------------------------------------------------------------|---------------------|
| 1      | Carbon%                    | 0.23                                                                                     |                     |
| 2      | Manganese%                 | 1.5                                                                                      |                     |
| 3      | Sulphur%                   | 0.045                                                                                    |                     |
| 4      | Phosphorous%               | 0.045                                                                                    |                     |
| 5      | Silicon%                   | 0.4                                                                                      |                     |
| 6      | Carbon equivalent %( Max.) | 0.42                                                                                     |                     |
| 7      | Mode of De-oxidation       | Semi-killed or killed                                                                    |                     |

Signature of the Bidder with Company Seal

\*\* This form is to be duly filled in by the Bidder and submitted along with the Tender

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